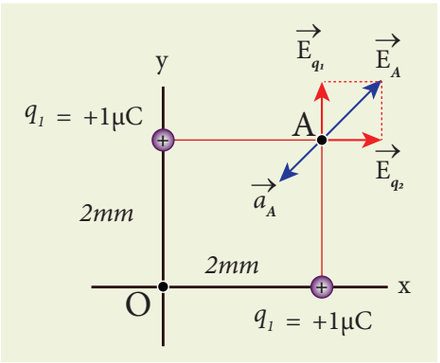
$$= 2.25\times10^{9}\ \hat{j} + 2.25\times10^{9}\hat{i} = 2.25\times10^{9}\left(\hat{i}+\hat{j}\right)$$

The magnitude of electric field

$$\left|\vec{E}_A\right| = \sqrt{\left(2.25\times10^{9}\right)^2 + \left(2.25\times10^{9}\right)^2}$$
$$= 2.25\times\sqrt{2}\times10^{9}\ \mathrm{NC}^{-1}$$

The direction of $\vec{E}_A$ is given by $\dfrac{\vec{E}_A}{\left|\vec{E}_A\right|} = \dfrac{2.25\times10^{9}\left(\hat{i}+\hat{j}\right)}{2.25\times\sqrt{2}\times10^{9}} = \dfrac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}$ which is the unit vector along OA as shown in the figure.



The acceleration experienced by an electron placed at point A is

$$\vec{a}_A = \frac{\vec{F}}{m} = \frac{q\vec{E}_A}{m}$$
$$= \frac{\left(-1.6\times10^{-19}\right)\times\left(2.25\times10^{9}\right)\left(\hat{i}+\hat{j}\right)}{9.1\times10^{-31}}$$
$$= -3.95\times10^{20}\left(\hat{i}+\hat{j}\right)\mathrm{N\ kg}^{-1}$$

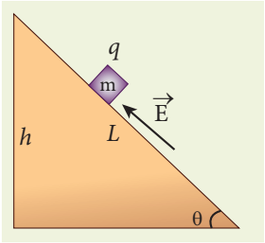
The electron is accelerated in a direction exactly opposite to $\vec{E}_A$.

1.3.3 Electric field due to continuous charge distribution

The electric charge is quantized microscopically. The expressions (1.2), (1.3), (1.4) are applicable to only point charges. While dealing with the electric field due to a charged sphere or a charged wire etc., it is very difficult to look at individual charges in these charged bodies. Therefore, it is assumed that charge is distributed continuously on the charged bodies and the discrete nature of charges is not considered here. The electric field due to such continuous charge distributions is found by invoking the method of calculus. (For further reading, refer Appendix A1.1).

EXAMPLE 1.8

A block of mass *m* carrying a positive charge *q* is placed on an insulated frictionless inclined plane as shown in the figure. A uniform electric field *E* is applied parallel to the inclined surface such that the block is at rest. Calculate the magnitude of the electric field *E*.



Solution

Note: A similar problem is solved in XI[th] Physics volume I, unit 3 section 3.3.2.
There are three forces that acts on the mass *m*:

(i) The downward gravitational force exerted by the Earth (*mg*)

(ii) The normal force exerted by the inclined surface (*N*)

(iii) The Coulomb force given by uniform electric field (*qE*)

The free body diagram for the mass *m* is drawn below.

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